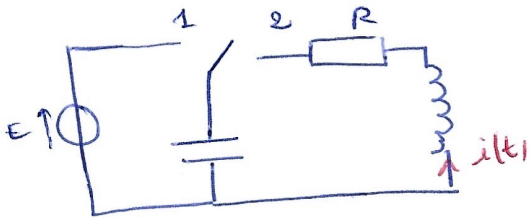
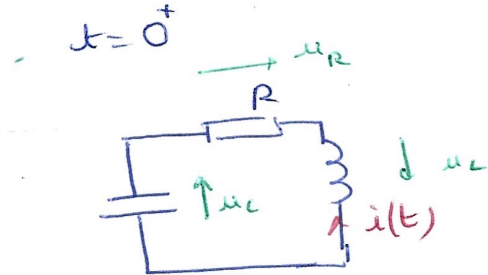
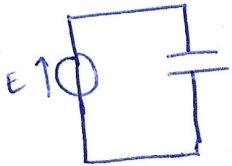


Circuit linéaire

d'ordre 2.



$t = 0^-$:



1) Loi des mailles :

$$u_C + u_R + u_L = 0$$

$$\frac{q(t)}{C} + R i(t) + L \frac{di(t)}{dt} = 0 \quad (*)$$

$$i = \frac{dq}{dt}$$

$$\frac{q}{C} + R \frac{dq}{dt} + L \frac{d^2 q}{dt^2} = 0$$

$$\frac{d^2 q}{dt^2} + \frac{R}{L} \frac{dq}{dt} + \frac{q}{LC} = 0$$

2)

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

2

$$\frac{\omega_0}{Q} = \frac{R}{L}$$

$$\frac{L}{R} \frac{1}{\sqrt{LC}} = Q = \frac{1}{R} \sqrt{\frac{L}{C}}$$

$$3) E_{\text{tot}} = \frac{1}{2} C u^2(t) + \frac{1}{2} L i^2(t).$$

$$E_{\text{tot}} = \frac{1}{2} \frac{q^2}{C} + \frac{1}{2} L i^2$$

4)

$$2) (*) \frac{q}{C} \cdot i + R i^2 + L i \frac{di}{dt} = 0$$

$$\frac{q}{C} \frac{dq}{dt} + R i^2 + \frac{d}{dt} \left(\frac{1}{2} L i^2 \right)$$

$$\frac{d}{dt} \left(\frac{1}{2} \frac{q^2}{C} \right) + R i^2 + \frac{d}{dt} \left(\frac{1}{2} L i^2 \right) = 0$$

$$\frac{d}{dt} (E_{\text{tot}}) = - R i^2$$

5)

1 L'énergie stockée diminue car 1 partie est dissipée par effet Joule de la résistance.

6) Il faut $\varphi > 1/2$.

$$\text{On a } \Omega = \omega_0 \sqrt{1 - \left(\frac{1}{2\varphi} \right)^2}$$

$$T = \frac{2\pi}{\Omega}$$

$$T_0 = \frac{2\pi}{\omega_0}$$

$$T = \frac{1}{\omega_0} \frac{2\pi}{\sqrt{1 - \left(\frac{1}{2\varphi} \right)^2}}$$

$$\left. \frac{1}{\sqrt{1 - \left(\frac{1}{2\varphi} \right)^2}} \right\} < 1 \quad \left. \right\} > 1$$

$$T_{pp} = \frac{1}{\omega_0} \frac{4\pi \cdot \varphi}{\sqrt{4\varphi^2 - 1}}$$

On a bien $T_{pp} > T_0$.

7] A $t=0^-$: condensateur chargé

$$q(0^-) = q_0 = CE$$

on a C^0 ds 1 cond et $u = \frac{q}{C}$ dc q et C^0 est

$$\Rightarrow q(0^+) = q_0 = CE$$

$\frac{dq}{dt}(0^+) = i(0^+) = i(0^-)$ car i continu ds L .

$$i(0^-) = 0 \text{ car circuit 2 ouvert} \Rightarrow \frac{dq}{dt}(0^+) = 0$$

8] On a

$$q(t) = e^{-\frac{\omega_0}{2Q}t} [C_1 \cos \Omega t + C_2 \sin \Omega t]$$

$$\frac{dq}{dt} = e^{-\frac{\omega_0}{2Q}t} \left[-\frac{\omega_0}{2Q} [C_1 \cos \Omega t + C_2 \sin \Omega t] - \Omega C_1 \sin \Omega t + C_2 \Omega \cos \Omega t \right]$$

$$\begin{cases} \frac{dq}{dt}(0) = -\frac{\omega_0}{2Q} C_1 + C_2 \Omega \\ q(0) = C_1 = CE \end{cases} \Rightarrow C_2 = \frac{\omega_0}{\Omega \cdot 2Q} C_1$$

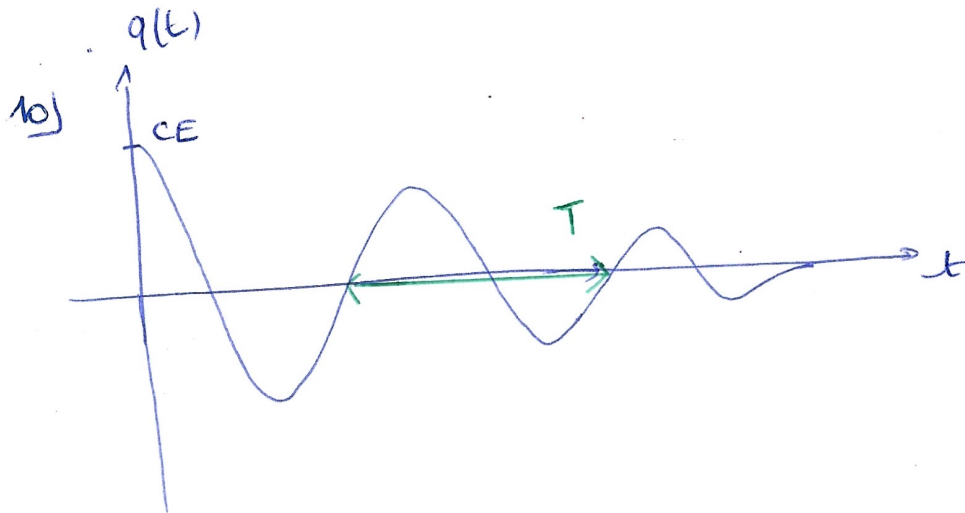
Finale :

$$q(t) = CE e^{-\frac{\omega_0}{2Q}t} \left[\cos \Omega t + \frac{\omega_0}{2Q\Omega} \sin \Omega t \right]$$

$$\frac{\omega_0}{\Omega \cdot 2Q} = \frac{1}{2Q \sqrt{1 - \left(\frac{1}{2Q}\right)^2}} = \frac{1}{\sqrt{4Q^2 - 1}}$$

$$q(t) = CE e^{-\frac{\omega_0}{2Q}t} \left[\cos \Omega t + \frac{1}{\sqrt{4Q^2 - 1}} \sin \Omega t \right]$$

9]

 $q = 0$ en régime permanent.(voir solution $q(t)$ sur condensateur déchargé).

Q: nb de périodes visibles sur l'oscillogramme.